Assignment- X

Group Project: Back End

DUE: 3rd November

Total Marks: 10

Submission Instructions

Each group may prepare a single common report. However, every team member must submit the same PDF individually on VTOP.
Include the following details for every team member:

- Name
- Registration Number
- Lab Slot (e.g., L9 + L10)

This submission documents your progress through the **back-end** stages of the ASIC design flow, from floor planning through physical verification. Your report should describe the physical implementation of your design, present the relevant results, and discuss the findings.

Implementation and Results

Submit a PDF report documenting your implementation and results. Include supporting screenshots, tool reports, and other relevant evidence.
The report should cover the following stages, as applicable:

- **Floorplanning:** Document the floorplan, core and die dimensions, aspect ratio, and target utilization. Include the floorplan screenshot and discuss the choices made.
- **Power Planning:** Show the power distribution network and explain the power rings, stripes, and connections used in your design, as applicable.
- **Placement:** Present the placement results, standard-cell utilization, and relevant timing or congestion reports. Discuss any issues encountered and the optimization performed.

- **Clock Tree Synthesis (CTS):** Document the clock tree and report relevant metrics.
- **Routing:** Present the routed design and discuss routing completion, congestion, and any relevant design-rule violations.
- **Timing Analysis and Closure:** Include post-placement, post-CTS, and/or post-route timing results, as applicable. Discuss setup and hold violations, timing improvements, and any remaining critical paths.
- **Physical Verification:** Present the results of Design Rule Checking (DRC) and Layout Versus Schematic (LVS), as applicable to the verification flow available to your group. Clearly identify any unresolved violations.

Discussion of Results

Your discussion should address:

- Whether the design meets its timing constraints.
- Whether the design is free of unresolved DRC and LVS violations, where these checks are applicable.
- The final area and other relevant implementation metrics.
- Any remaining limitations or violations, with an explanation of t

Include sufficient evidence to support your claims. Screenshots should be legible.

Individual Contributions

Clearly describe the contribution of each team member to the back-end implementation. Specify the tasks performed by each member and the work completed.